EXPERIMENT 1: Installing Anaconda/Miniconda & Setting Up TensorFlow & Keras {#experiment-1}

Objectives

- Install and configure Anaconda/Miniconda package manager
- Set up Python environment for deep learning
- Install TensorFlow and Keras
- Verify installation and run basic test

Prerequisites

- Windows/Linux/macOS machine
- Internet connection
- 5+ GB disk space

Implementation Steps

Step 1: Download and Install Anaconda/Miniconda

```
# For Miniconda (lightweight, recommended)
# Download from: https://docs.conda.io/en/latest/miniconda.html

# On Linux/macOS:
bash Miniconda3-latest-Linux-x86_64.sh

# On Windows:
# Run the .exe installer
```

Step 2: Create a Virtual Environment

```
# Create environment named "dl_env"
conda create -n dl_env python=3.10

# Activate environment
conda activate dl_env
```

Step 3: Install TensorFlow and Keras

```
# Install TensorFlow (includes Keras)
pip install tensorflow tensorflow-gpu

# Verify TensorFlow installation
python -c "import tensorflow as tf; print(tf.__version__)"
```

Step 4: Install Additional Libraries

```
pip install numpy pandas matplotlib scikit-learn jupyter notebook
```

Step 5: Verification Script

```
import tensorflow as tf
import keras
import numpy as np

print(f"TensorFlow Version: {tf.__version__}")
print(f"Keras Version: {keras.__version__}")
print(f"GPU Available: {tf.config.list_physical_devices('GPU')}")

# Test basic operation
a = tf.constant([[1., 2.], [3., 4.]])
print(f"Matrix multiplication: {tf.matmul(a, a)}")
```

Expected Outcomes

- Successfully installed Anaconda/Miniconda
- Virtual environment created and activated
- TensorFlow and Keras running without errors
- GPU detected (if available)

Common Issues & Solutions

Issue	Solution
CUDA not found	Reinstall tensorflow-gpu or use CPU version
Python version mismatch	Use Python 3.8-3.10
Permission denied	Use `sudo` or conda for installation